Their gliders kept failing. So they tested 200+ wings.

Two gliders. Not enough lift.

1900 glider: short on lift

1901 glider: short on lift

1901 **wind tunnel**

They trusted Otto Lilienthal's tables.

> The best published numbers of the day.

A wooden wind tunnel in a bicycle shop.

🚲 Bicycle shop

💨 Fan + tunnel

📏 Test wings

Preliminary tests on more than 200 wing shapes.

200+

wing shapes tested

THEN

The best ones, studied in detail.

The data was right, but only for one wing shape.

LILIENTHAL'S WING

Data fits

ANY OTHER WING

Data misleads

Dec 17, 1903: 12 seconds in the air.

12 s

Orville's first powered flight

When the numbers don't work, test your own case.

> Save this for when you're stuck.